

Centre Number	Candidate Number	Candidate Name
---------------	------------------	----------------

NAMIBIA SENIOR SECONDARY CERTIFICATE

BIOLOGY ADVANCED SUBSIDIARY LEVEL

8223/3

PAPER 3 Advanced Practical Skills

2 hours

Marks 40

2023

Additional Materials: As listed in Instructions to subject teachers.

INSTRUCTIONS AND INFORMATION TO CANDIDATES

- Candidates answer on the question paper in the spaces provided.
- Write your Centre Number, Candidate Number and Name in the spaces at the top of this page.
- Write in dark blue or black pen.
- You may use a soft pencil for any rough work, diagrams or graphs.
- Do not use correction fluid.
- You may use a non-programmable calculator.
- Do not write in the margin *For Examiner's Use*.
- Answer **all** questions.
- The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	
2	
Total	

Marker	
Checker	

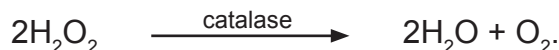
This document consists of **10** printed pages and **2** blank pages.



Republic of Namibia
MINISTRY OF EDUCATION, ARTS AND CULTURE

Before you proceed, read carefully through the whole of Question 1 and Question 2. Plan the use of the **two hours** to make sure that you finish the whole of Question 1 and Question 2.

- 1 Catalase is an enzyme found in nearly all living organisms. It is involved in the breakdown of hydrogen peroxide into water and oxygen, as shown by the equation below:



In this experiment you are going to observe the effect that the substrate concentration has on enzyme activity. For that purpose, you need to carry out a serial dilution of the hydrogen peroxide **P**, to halve its concentration between each successive dilution.

Fig. 1.1 shows how to make the first dilution **Q**.

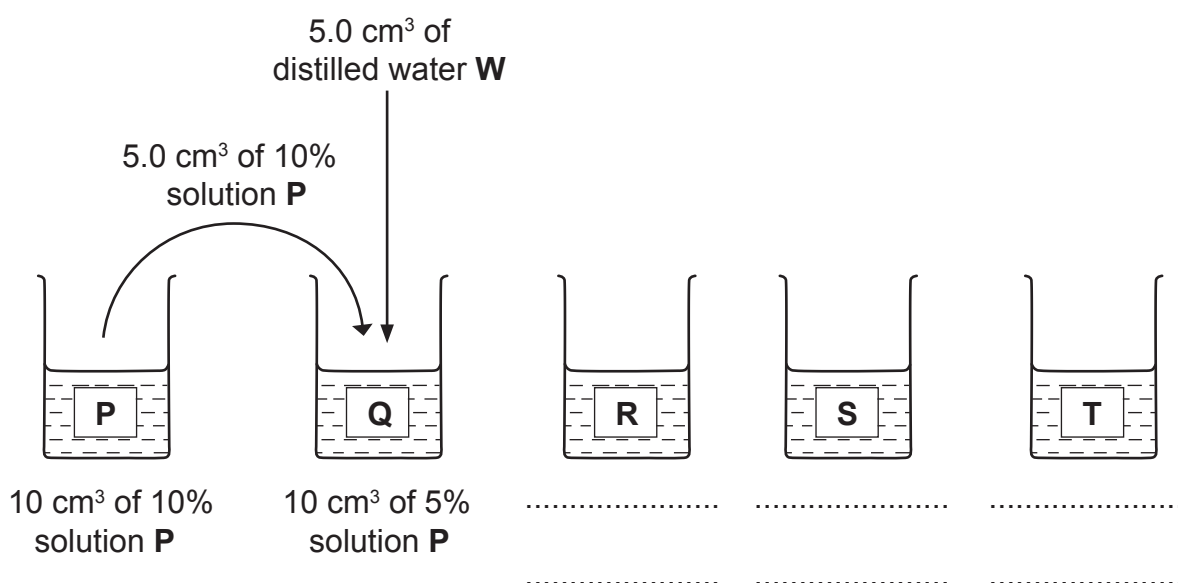


Fig. 1.1

- (a) Complete Fig.1.1 to show how you would make three further serial dilution **R**, **S** and **T** of the original hydrogen peroxide **P**. [3]
Now proceed to make solution **Q**, **R**, **S** and **T** according to the method described above in the beakers. Once you are done making the solutions, add the contents of the beakers to the test-tubes with corresponding letter e.g. contents of **beaker P** is added to **test-tube P**.

- (b)** To determine the rate of the enzyme reaction, you will add immobilized catalase to the substrate and time how long it takes before the bead reaches the surface of the hydrogen peroxide solution. Carefully read through the steps below before you start with the experiment.

Step 1: Take **one** of the immobilized catalase beads from the Petri dish labelled **beads** and drop it into the test-tube labelled **P**.

Step 2: As soon as you add the bead, start your stopwatch.

Step 3: The bead will sink to the bottom, after which it will rise to the surface of the solution again. Stop the time when the bead reaches the surface of the solution.

Step 4: Take a new bead and add to beaker **P** another **two** times, each time measuring how long it takes to reach the surface. Record your results in the table you prepare at **(b)(i)**.

Step 5: Repeat Steps 1 to 4 for solutions **Q**, **R**, **S** and **T**.

- (i)** Prepare the space below to record your observations from Step 3, Step 4 and Step 5. The table should also include the % concentration of the solution, as well as the average time it took the bead to rise in each solution.

[4]

- (ii) Describe the correlation between hydrogen peroxide concentration and the average time taken for the catalase beads to rise to the surface. In your answer try to establish a link between one of the products formed and the rising of the beads.

.....

.....

.....

.....

.....

.....

[3]

- (iii) Identify **one** significant source of error when determining the time taken for the catalase beads to rise to the surface in Step 2 and 3.

.....

.....

[1]

- (c) To make the experiment fair, scientists need to ensure that various variables remain constant.
State **two** variables other than pH that need to remain constant when performing the experiment.
Describe a method that could be used to ensure that those variables remain constant.

variable 1

description.....

.....

.....

.....

variable 2

description.....

.....

.....

.....

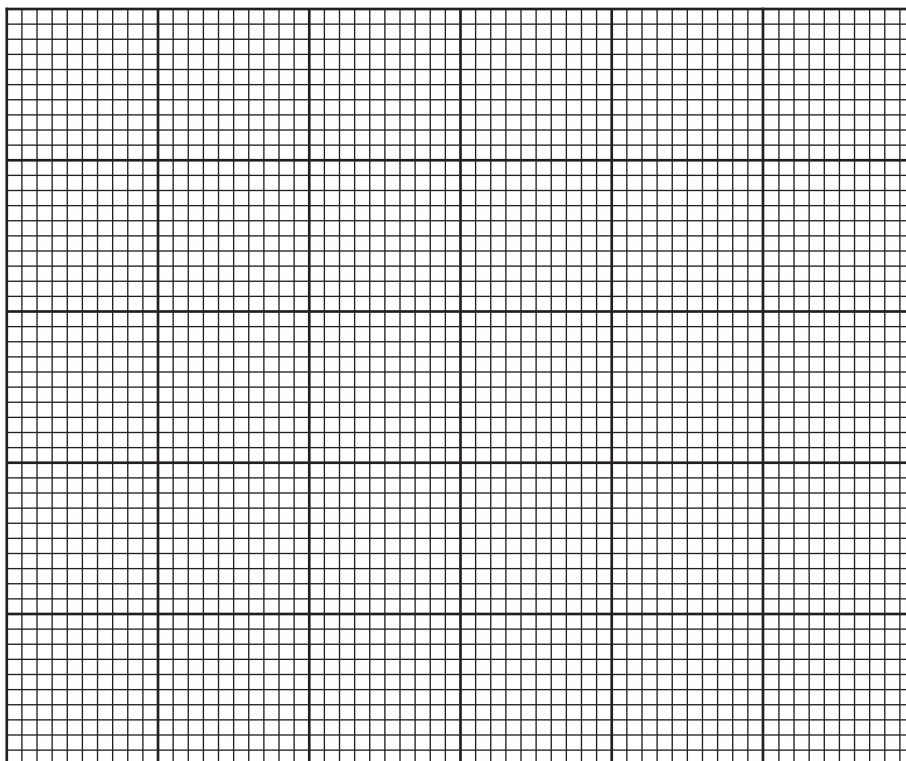
[4]

- (d) In a similar experiment another group of students measured the effect that temperature had on the rate of reaction. The students changed the temperature, while keeping hydrogen peroxide concentration, pH, and all other variables the same. The average time it took the catalase bead to rise to the surface was recorded in Table 1.1.

Table 1.1

temperature / °C	average time taken for bead to rise to surface / s
29	19
32	13
35	8
38	4
42	12
46	26
51	44

- (i) Plot a graph of the data shown in Table 1.1.



- (ii) Use your graph to estimate the temperature, at which it will take the beads 40s to reach the surface. Clearly indicate on the graph how you determined your answer. [4]

Temperature = °C [1]

[Total: 20]

- 2** You are provided with a sample of onion kept in solution **N1**. You are required to make a slide of the purple/red epidermis of the onion as follows:
- Place a drop of solution **N1** onto the microscope slide.
 - Peel off a piece of the **purple/red epidermis** of the onion using forceps or your fingers, as indicated in Fig. 2.1.



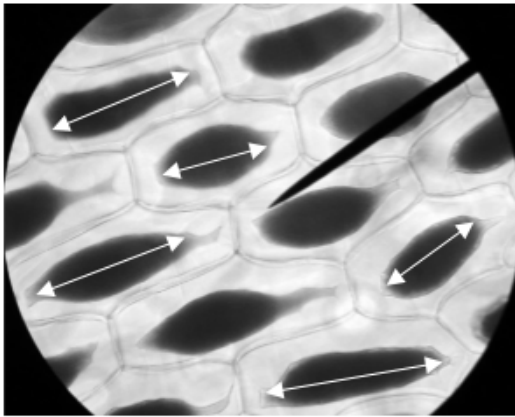
Fig. 2.1

- Cut a piece of epidermis small enough to fit under the cover slip.
 - Place the small piece of epidermis into the drop of **N1** solution and ensure that enough solution is present to prevent the epidermis from drying out.
 - Cover the epidermis with a cover slip and remove any excess solution using the paper towel supplied.
- (a)** Use the microscope and select an area of cells that shows the effect of keeping the cells in solution **N1**.

Draw **three** adjacent cells and label **two visible** organelles.

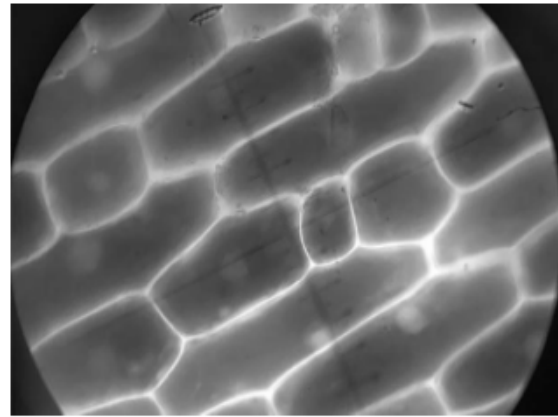
(b) Similarly to the piece of onion being kept in solution **N1**, two more pieces were kept in solution **N2** and **N3** respectively.

- Fig. 2.2 shows a photomicrograph of onion cells from solution **N2**.
- Fig. 2.3 shows a photomicrograph of onion cells from solution **N3**.



Magnification $\times 400$

Fig. 2.2



Magnification $\times 400$

Fig. 2.3

- (i) State which of the solutions, **N2** or **N3** has the lower water potential. Give a reason for your answer.

.....

[1]

- (ii) To find the average diameter of the cytoplasm in onion cells placed in solution **N2**, a student measured the diameter of the cytoplasm of five cells as indicated by the white arrows in Fig. 2.2.

Use the magnification to calculate the average actual diameter, in μm , of the cytoplasm using the five arrows. Measure the length of the arrows from tip to tip. Show your working.

..... μm

[4]

- (iii) Prepare a table in the space below to show the **visible differences** between the specimens in Fig. 2.2 and Fig. 2.3. Record your observations in the table you have prepared, also clearly indicating the feature you refer to.

[4]

- (c) Plant cells are not only sensitive to the composition of their surroundings, but also to the temperature of their environment. In an investigation the permeability of the beetroot cell membrane was tested at different temperatures. The colour of beetroot is due to the presence of a red pigment called anthocyanin

Three same sized discs of the same beetroot were placed into each of 5 identical test-tubes. To each test-tube the same volume of water was added. The test-tubes were then heated for 30 minutes at 30°C, 40°C, 50°C, 60°C and 70°C respectively.

After the 30 minutes, water from each test-tube was taken and its absorbency tested with a colorimeter. Fig. 2.4 shows the test-tubes at their respective temperatures and with their respective colours, whereas Table 2.1 shows the results from the colorimeter.

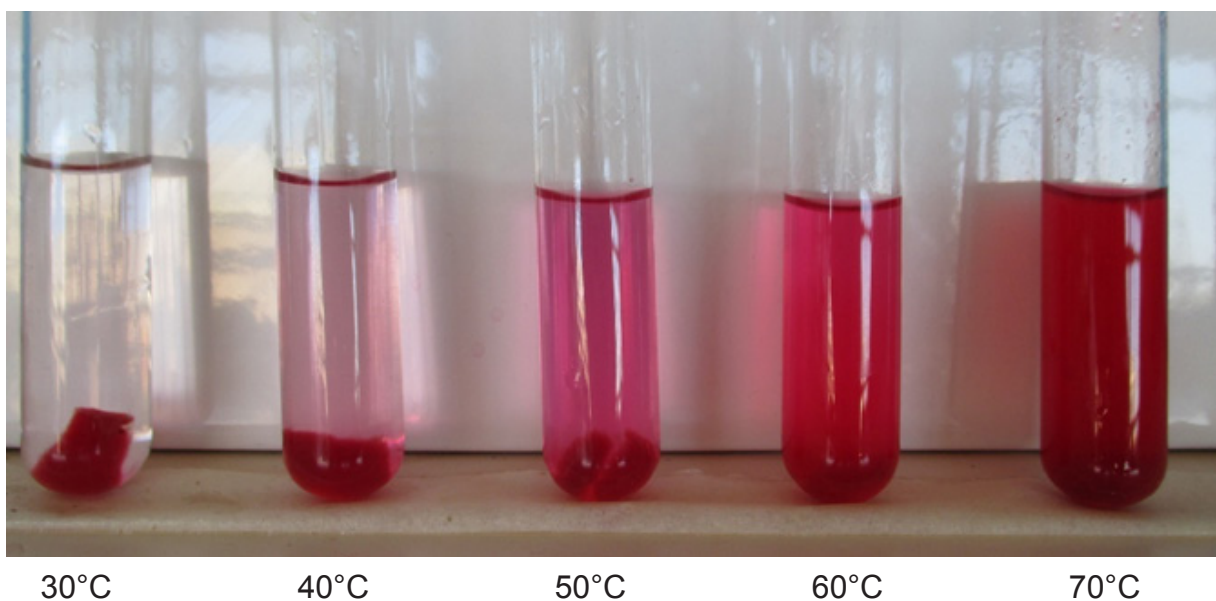


Fig. 2.4

Table 2.1

temperature / °C	absorbance of light / a.u
30	0.11
40	0.17
50	0.31
60	0.87
70	1.66

- (i) On the system of axes supplied, sketch a graph to show the effect that temperature has on the absorbency of light.



[2]

- (ii) Describe the relationship between temperature and the permeability of cell membranes.

.....

[1]

- (iii) During the experiment, several variables were kept constant. These included:

- beetroot discs of the same size
- identical test-tubes
- same volume of water
- beetroot discs heated for 30 minutes

State **two** other factors that should be kept the same during the experiment.

1

2

[2]

(iv) Suggest **two** improvements which can be made to the investigation with the beetroot in order to make the results more reliable.

1

.....

2

.....

[2]

[Total: 20]

BLANK PAGE

BLANK PAGE

The DNEA acknowledges the usage and reproduction of third party copyright material in the NSSC Assessment, **with and without permission** from the copyright holder. The Namibian Government Copyright Act allows copyright material to be used limitedly and fairly for educational and non-commercial purposes.

The Directorate of National Assessment and Examinations operates under the auspices of the Ministry of Education, Arts and Culture in Namibia.